14 REACTIVITY SERIES

The reactivity series lists elements (mainly metals) in order of decreasing reactivity. It is likely that you will have come across some of this chemistry already in earlier years. This chapter looks at some of the reactions of metals and introduces the idea of redox reactions.



▲ Figure 14.1 Rails are welded together using molten iron, produced by a reaction between aluminium and iron(III) oxide.



▲ Figure 14.2 Gold is so unreactive that it will remain chemically unchanged in contact with air or water basically forever.

LEARNING OBJECTIVES

- Understand how metals can be arranged in a reactivity series based on their reactions with:
  - water
  - dilute hydrochloric or sulfuric acid
- Understand how metals can be arranged in a reactivity series based on their displacement reactions between:
  - metals and metal oxides
  - metals and aqueous solutions of metal salts
- Know the order of reactivity of these metals: potassium, sodium, lithium, calcium, magnesium, aluminium, zinc, iron, copper, silver, gold
- Know the conditions under which iron rusts
- Understand how the rusting of iron may be prevented by
  - barrier methods
  - galvanising
  - sacrificial protection
- Understand the terms:
  - oxidation
  - reduction
  - redox
  - oxidising agent
  - reducing agent

  in terms of gain or loss of oxygen and loss or gain of electrons
- Practical: Investigate reactions between dilute hydrochloric and sulfuric acids and metals (e.g. magnesium, zinc and iron)

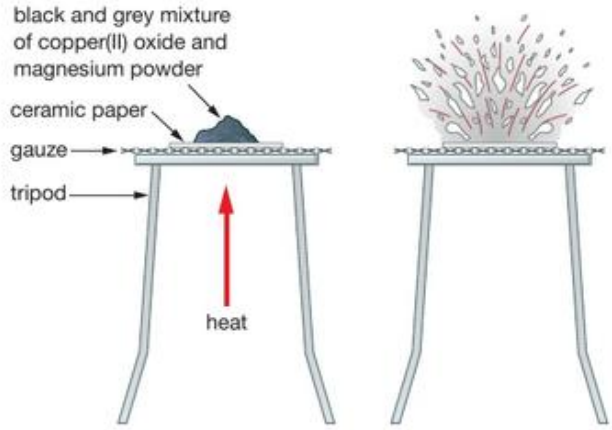
DISPLACEMENT REACTIONS INVOLVING METAL OXIDES

THE REACTION BETWEEN MAGNESIUM AND COPPER(II) OXIDE

potassium
sodium
lithium
calcium
magnesium
aluminium
(carbon)
zinc
iron
(hydrogen)
copper
silver
gold

decreasing reactivity

▲ Figure 14.4 The reactivity series. Although carbon and hydrogen are not metals they are usually included in the series. You will learn why in this chapter.



▲ Figure 14.3 Magnesium reacts with copper(II) oxide.

!

Safety Note: For this demonstration goggles or a face shield are needed, together with safety screens in a well-ventilated room. The bench should be protected by a large sheet of hardboard. The pupils should be at the back of the room. One gram of the mixture is strongly heated in a metal crown bottle cap before retiring to the back of the room to await a violent reaction. If nothing happens after a minute or so turn off the gas and do not touch the mixture but leave it to cool and dispose of it in water.

Magnesium powder and copper(II) oxide are mixed together and heated very strongly. At the end, traces of white magnesium oxide are left on the ceramic paper.

magnesium + copper(II) oxide → magnesium oxide + copper

$Mg(s) + CuO(s) \rightarrow MgO(s) + Cu(s)$

This is an example of a displacement reaction. The less reactive metal, copper, has been displaced from its compound by the more reactive magnesium. Any metal higher in the reactivity series will displace one lower down from a compound.

If you heated copper with magnesium oxide, nothing would happen because copper is less reactive than magnesium. Copper isn't capable of displacing magnesium from magnesium oxide.

KEY POINT

These reactions involving metals and metal oxides are often called *competition reactions*.

THE REACTION BETWEEN MAGNESIUM AND ZINC OXIDE

Heating magnesium with zinc oxide produces zinc metal:

magnesium + zinc oxide → magnesium oxide + zinc

$Mg(s) + ZnO(s) \rightarrow MgO(s) + Zn(s)$

THE REACTION BETWEEN CARBON AND COPPER(II) OXIDE

!

Safety Note: Wear eye protection. The tube will get very hot and remain so for some time.

A black mixture of carbon and copper(II) oxide is heated in a test-tube. The mixture glows red hot because of the heat given out during the reaction, and you are left with pink-brown copper in the tube.

$C(s) + 2CuO(s) \rightarrow CO_2(g) + 2Cu(s)$

Carbon is above copper in the reactivity series and displaces the copper from copper(II) oxide.

DID YOU KNOW?

Many metals are found as metal oxides in nature. If they are found as other compounds these are often converted to metal oxides before the metal is extracted.

CHEMISTRY ONLY

Carbon is included in the reactivity series because it is important in extracting several metals (including iron) from metal oxides. If the metal is less reactive than carbon (below carbon in the reactivity series), then heating with carbon can be a cheap way of removing oxygen from the oxide to leave the metal. Copper isn't, in fact, extracted like this. This reaction is simply a lab illustration that carbon is above copper in the reactivity series.

END OF CHEMISTRY ONLY

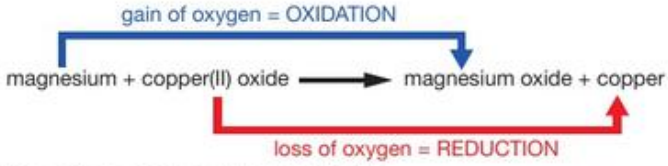
OXIDATION AND REDUCTION

OXYGEN TRANSFER

A substance has been oxidised if it gains oxygen. **Oxidation** is gain of oxygen.

A substance has been reduced if it loses oxygen. **Reduction** is loss of oxygen.

Consider the reaction between magnesium and copper(II) oxide again. Figure 14.5 shows what happens in terms of oxidation and reduction.



▲ Figure 14.5 Magnesium reacting with copper(II) oxide

A **redox** reaction is one in which both **red**uction and **ox**idation are occurring. Oxidation and reduction always occur together because if something loses oxygen, something else must gain it.

- A **reducing agent** is a substance that reduces something else. In this case, the magnesium is the reducing agent because it takes the oxygen away from the copper oxide, reducing it.
- An **oxidising agent** is a substance that oxidises something else. The copper(II) oxide is the oxidising agent in this reaction because it gives oxygen to the magnesium, oxidising it.

An oxidising agent always gets reduced in a chemical reaction because it oxidises something else by giving away its oxygen.

A reducing agent always gets oxidised in a chemical reaction because it takes the oxygen away from something else and therefore gains oxygen itself.

In the reaction between magnesium and zinc oxide on page 146, the magnesium is the reducing agent and the zinc oxide is the oxidising agent. In the reaction between copper(II) oxide and carbon, the carbon is the reducing agent and the copper(II) oxide is the oxidising agent.

ELECTRON TRANSFER

We are now going to look very closely at what happens in the reaction between magnesium and copper(II) oxide in terms of the various particles involved. Here is the equation again:

$$Mg(s) + CuO(s) \rightarrow MgO(s) + Cu(s)$$

The magnesium and the copper are metals, and are made of metal atoms, but the copper(II) oxide and the magnesium oxide are both ionic compounds.

The copper(II) oxide contains Cu^{2+} and O^{2-} ions, and the magnesium oxide contains Mg^{2+} and O^{2-} ions. Writing these into the equation gives:

$$Mg(s) + Cu^{2+}O^{2-}(s) \rightarrow Mg^{2+}O^{2-}(s) + Cu(s)$$

The oxide ion (O^{2-}) does not change in this reaction. It ends up with a different partner, but is totally unchanged itself.

What is happening in this reaction is that magnesium *atoms* are turning into magnesium *ions*. The magnesium atoms lose electrons to form magnesium ions. These electrons are gained by the copper ions to form copper atoms. We have described this as a redox reaction above in terms of loss/gain of oxygen, but now we can see that it was not the oxygen that was actually the important part, it has not changed! To understand what is happening better, we need a more fundamental definition of oxidation and reduction:

- *oxidation is loss of electrons*
- *reduction is gain of electrons.*

The mnemonic OILRIG can be helpful in remembering this:

Oxidation

Is

Loss of electrons

Reduction

Is

Gain of electrons

DISPLACEMENT REACTIONS INVOLVING SOLUTIONS OF SALTS

Salts are compounds such as copper(II) sulfate, silver nitrate or sodium chloride. You will find a definition of what a salt is on page 173. This section explores some reactions between metals and solutions of salts in water.

THE REACTION BETWEEN ZINC AND COPPER(II) SULFATE SOLUTION

The copper is displaced by the more reactive zinc. The blue colour of the copper(II) sulfate solution fades as colourless zinc sulfate solution is formed:

$$Zn(s) + CuSO_4(aq) \rightarrow ZnSO_4(aq) + Cu(s)$$

The zinc and the copper are metals consisting simply of atoms, but the copper(II) sulfate and the zinc sulfate are metal compounds and so are ionic.

The equation can be rewritten showing the ions:

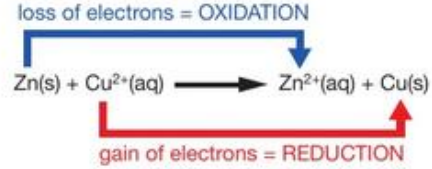
$$Zn(s) + Cu^{2+}(aq) + SO_4^{2-}(aq) \rightarrow Zn^{2+}(aq) + SO_4^{2-}(aq) + Cu(s)$$

We can see that the sulfate ions are exactly the same on both sides of the equation – they have not changed at all in the reaction. We call the sulfate ions here *spectator ions*. Removing the spectator ions (because they aren't changed during the reaction) leaves:

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

This is called an ionic equation and just shows the things that change in the reaction.

This is another redox reaction.



▲ Figure 14.6 The reaction between zinc and copper(II) sulfate solution

KEY POINT

When we looked at the reaction between Mg and CuO above, we also wrote ions in the equation but we did not separate them completely because the ions are in a solid. We only separate everything like this when things are in solution. In a solution the ions are free to move around separately, in a solid they are not.

KEY POINT

Which copper(II) salt you started with would not matter, as long as it was soluble in water. Copper(II) chloride or copper(II) nitrate would react in exactly the same way with zinc because the chloride ions or the nitrate ions would again be spectator ions, not participating in the reaction.

The Zn atoms are oxidised to Zn^{2+} ions because they lose electrons. The Cu^{2+} ions gain electrons and are reduced to Cu atoms. We can split up the ionic equation to show the individual oxidation and reduction processes:

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^- \quad \text{oxidation}$$

$$Cu^{2+}(aq) + 2e^- \rightarrow Cu(s) \quad \text{reduction}$$

These are called *ionic half-equations* as they each only show one of the processes (either oxidation or reduction) occurring in the reaction. In reality, these processes cannot occur one without the other: if something gains electrons, it has to get them from somewhere, so something else must lose electrons.

OXIDISING AND REDUCING AGENTS IN TERMS OF ELECTRONS

A couple more definitions (this time in terms of electrons):

- *An oxidising agent is something that oxidises something else by taking electrons away from it. Oxidising agents accept electrons and therefore are reduced in a reaction.*
- *A reducing agent is something that reduces something else by giving electrons to it. Reducing agents give away electrons and therefore are oxidised in a reaction.*

HINT

You now have two definitions of oxidation (and its reverse, reduction). Which one should you use? Use whichever is simpler in the case you are asked about! Both definitions are true. Don't worry too much about this at the moment. With a little experience, you will find it is obvious which one you need to use in various cases.

Let us look again at the overall ionic equation for the displacement reaction between zinc and copper ions:

$$Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$$

The Zn reduces the Cu^{2+} by giving electrons to it and therefore the Zn is the reducing agent. In the process of the reaction the Zn is oxidised because it has given electrons away (lost them).

The Cu^{2+} oxidises the Zn by taking electrons away from it, therefore the Cu^{2+} is the oxidising agent. In the process the Cu^{2+} is reduced because it has taken electrons (gained them).



▲ Figure 14.7 **Displacing silver from silver nitrate solution**

THE REACTION BETWEEN COPPER AND SILVER NITRATE SOLUTION

Silver is below copper in the reactivity series, so a coil of copper wire in silver nitrate solution will produce metallic silver. Figure 14.7 shows the silver being produced as a mixture of grey 'fur' and delicate crystals. Notice the solution becoming blue, as copper(II) nitrate is produced:

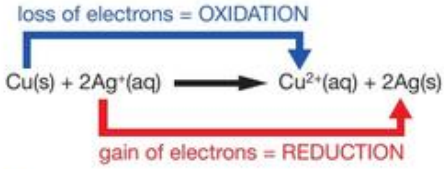
$$Cu(s) + 2AgNO_3(aq) \rightarrow Cu(NO_3)_2(aq) + 2Ag(s)$$

This time the nitrate ions are spectator ions, and the final version of the ionic equation looks like this:

$$Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$$

This is another redox reaction.

The Cu is the reducing agent (it is oxidised) because it gives electrons to the Ag^+ ions to reduce them to Ag. The Ag^+ ion is the oxidising agent (it is reduced) because it takes electrons away from the Cu atoms.



▲ Figure 14.8 Copper reacting with silver nitrate solution

The ionic half-equations are:

$Cu(s) \rightarrow Cu^{2+}(aq) + 2e^-$ oxidation

$Ag^+(aq) + e^- \rightarrow Ag(s)$ reduction

REACTIONS OF METALS WITH WATER

A GENERAL SUMMARY

METALS ABOVE HYDROGEN IN THE REACTIVITY SERIES

Metals above hydrogen in the reactivity series react with water (or steam) to produce hydrogen.

If the metal reacts with *cold water*, the metal *hydroxide* and hydrogen are formed.

metal + cold water → metal hydroxide + hydrogen

If the metal reacts with *steam*, the metal *oxide* and hydrogen are formed.

metal + steam → metal oxide + hydrogen

As you move down the reactivity series, the reactions become less and less vigorous.

METALS BELOW HYDROGEN IN THE REACTIVITY SERIES

Metals below hydrogen in the reactivity series (such as copper) don't react with water or steam. This is why copper can be used for both hot and cold water pipes.

REACTIONS OF POTASSIUM, SODIUM OR LITHIUM WITH COLD WATER

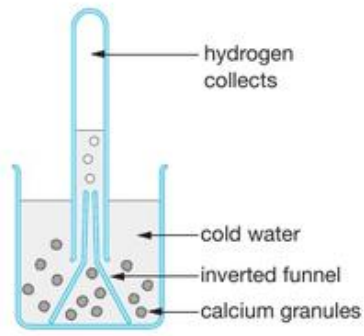
These reactions are described in detail on pages 125–126. They are very vigorous reactions, but become less violent in the following order: potassium > sodium > lithium. The equations all look like this:

$2M(s) + 2H_2O(l) \rightarrow 2MOH(aq) + H_2(g)$

Replace M by K, Na or Li, depending on which metal you want.

REACTION OF CALCIUM WITH COLD WATER

Calcium reacts gently with cold water. The grey granules sink, but are carried back to the surface again as bubbles of hydrogen are formed around them. The mixture becomes warm as heat is produced.



▲ Figure 14.9 Calcium reacting with cold water

Calcium hydroxide is formed. This isn't very soluble in water. Some of it dissolves to give a colourless solution, but most of it is left as a white, insoluble solid.

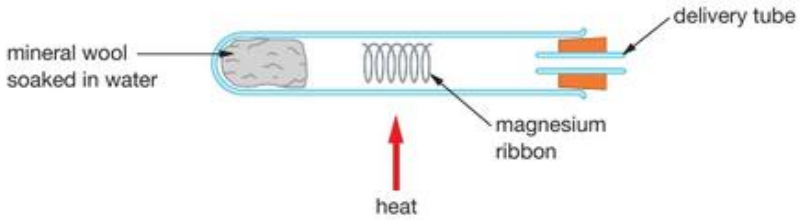
$Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq \text{ or } s) + H_2(g)$

REACTION OF MAGNESIUM WITH COLD WATER

There is almost no reaction. If the magnesium is very clean, a few bubbles of hydrogen form on it, but the reaction soon stops again. This is because the magnesium becomes coated with insoluble magnesium hydroxide, which prevents any more water coming into contact with the magnesium.

REACTION OF MAGNESIUM WITH STEAM

Magnesium ribbon can be heated in steam using the apparatus shown in Figure 14.10.



▲ Figure 14.10 Magnesium reacting with steam

The mineral wool isn't heated directly. Enough heat moves back along the test-tube to turn the water to steam.

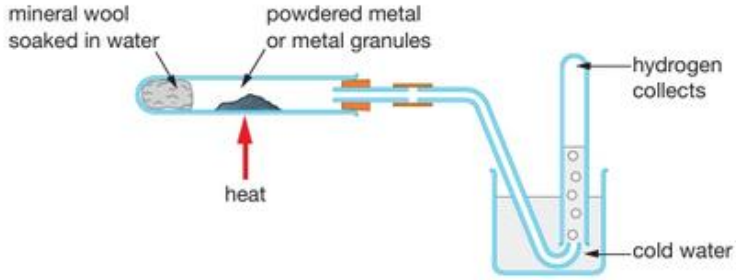
The magnesium burns with a bright white flame in the steam, producing hydrogen, which can be ignited at the end of the delivery tube. White magnesium oxide is formed:

$$Mg(s) + H_2O(g) \rightarrow MgO(s) + H_2(g)$$

KEY POINT

You might wonder why there is no description for aluminium and steam. The reactivity of aluminium is supposed to be between that of magnesium and that of zinc. However, aluminium has only a very slow reaction with steam because it is covered in a very thin, but very strong, layer of aluminium oxide. It only really shows its true reactivity if that layer can be penetrated in some way. Water or steam don't do that very well. We will talk about this again when we look at reactions between metals and acids.

REACTIONS OF ZINC OR IRON WITH STEAM



▲ Figure 14.11 Zinc or iron reacting with steam

With both zinc and iron, the hydrogen comes off slowly enough to be collected. Neither metal burns.

KEY POINT

Care has to be taken during this experiment to avoid 'suck-back'. If you stop heating while the delivery tube is still under the surface of the water, water is sucked back into the hot tube, which often results in it cracking.

WITH ZINC

Zinc oxide is formed. This is yellow when it is hot, but white on cooling.

$$Zn(s) + H_2O(g) \rightarrow ZnO(s) + H_2(g)$$

WITH IRON

The iron becomes slightly darker grey. A complicated oxide is formed, called tri-iron tetroxide, Fe_3O_4:

$$3Fe(s) + 4H_2O(g) \rightarrow Fe_3O_4(s) + 4H_2(g)$$

KEY POINT

Notice that in these equations, water now has a state symbol (g) because we are talking about it as steam.

REMINDER

Remember that *metals below hydrogen* in the reactivity series, such as copper, *don't react* with water or steam.

REACTIONS OF METALS WITH DILUTE ACIDS

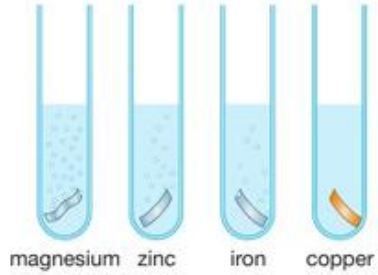
ACTIVITY 4

▼ PRACTICAL: INVESTIGATION INTO THE REACTIONS BETWEEN METALS AND DILUTE ACIDS

! Safety Note: Wear eye protection. Keep hands and face away from the mouths of the tubes as some acid aerosol (tiny droplets) may be carried out by the gas.

- Set up four test-tubes and put about 2 cm^3 of dilute hydrochloric acid into each one.
- Put a small piece of magnesium, zinc, iron or copper into each test-tube and observe any reaction that occurs.
- If there is fizzing, collect or trap the gas and test with a lighted splint – a squeaky pop indicates the presence of hydrogen gas.
- Repeat the experiments with dilute sulfuric acid.

The results that we could obtain from the experiments with dilute hydrochloric acid are shown in Figure 14.12 and Table 14.1.



▲ Figure 14.12 The reactions between metals and hydrochloric acid

Table 14.1 The results of experiments on the reaction between hydrochloric acid and some metals

Metal	Reaction with dilute hydrochloric acid
Magnesium	Reacts vigorously with lots of fizzing. The gas produced gave a squeaky pop with a lighted splint. A colourless solution is formed. The test-tube gets hot.
Zinc	Steady reaction. Fizzing. Enough gas eventually collected to produce a squeaky pop with a lighted splint. A colourless solution formed. The test-tube gets warmer.
Iron	Slow fizzing. Very little gas was collected in the time available. A very pale green solution formed. The test-tube got slightly warmer.
Copper	No change.

These reactions are all **exothermic** (they give out heat). See Chapter 19 for more about exothermic reactions.

Now that we have noticed that there is a temperature change in the reaction we could measure this temperature change to compare the reactivity of the metals. However, in order to do this we will have to think more carefully about the things that we have to keep the same in order to make this a valid (fair) test.

The volume and concentration of the hydrochloric acid are perhaps the most obvious things that we have to keep the same, but what about the mass of the metal? This is a more difficult question. If we use 0.24 g of metal in each experiment, we can work out that this is 0.24/24 = 0.010 mol magnesium but 0.24/56 = 0.0040 mol iron.

KEY POINT

If the reaction occurs very slowly, the heat will be given out over a long period of time. As the heat is being given out it will also be lost to the surroundings and the temperature change will be quite low. If the reaction occurs more quickly, the heat is given out more quickly to the solution, and there is less time for heat to be lost to the surroundings – we obtain a larger temperature change. A large surface area causes a faster rate of reaction, so we should use metal powders in these reactions. These things are discussed more in Chapters 19 and 20.

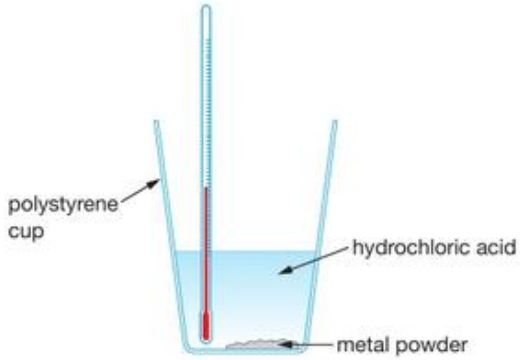
Now, chemical reactions depend on how many particles are present: more moles and therefore more particles are present in the Mg experiment and more heat would be given out because of this. It is therefore important that we use the same number of moles in each experiment rather than the same number of grams.

The other variable that we should control is the surface area of the metal. Although this does not affect the overall amount of heat given out in the reaction, it will affect the speed at which the heat is given out, which can affect the temperature change.

We will use the following procedure to investigate this reaction quantitatively (involving numbers):

- Measure out 50 cm^3 of 1 mol/dm^3 hydrochloric acid using a 50 cm^3 measuring cylinder (this amount was chosen so that the hydrochloric acid is in excess – there is more than enough to react with all the metal in each case).
- Pour the hydrochloric acid into a polystyrene cup (this is an insulator).
- Measure the initial temperature of the hydrochloric acid.
- Weigh out 0.010 mol of magnesium (0.010 × 24 = 0.24 g) powder.
- Add the magnesium powder to the polystyrene cup, stirring rapidly, and measure the maximum temperature reached.
- Repeat with the other metals using 0.010 mol in each case.

! Safety Note: Wear eye protection and avoid splashing the acid when stirring.

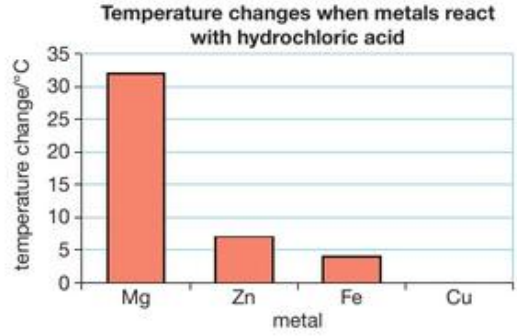


▲ Figure 14.13 Measuring the temperature change when a metal reacts with dilute hydrochloric acid.

The data for this experiment could be:

Metal	Mass/g	Initial temperature/°C	Maximum temperature/°C
magnesium	0.24	18	40
zinc	0.65	19	26
iron	0.56	18	22
copper	0.64	18	18

We can work out the temperature change for each experiment and plot this as a bar chart. We use a bar chart because the type of metal is not continuous data: the metal can be either magnesium or zinc, it cannot be anything in between.



▲ Figure 14.14 Magnesium gives out the most heat when reacted with hydrochloric acid.

We can see from this data that magnesium caused the greatest temperature change and so is the most reactive metal of the four; copper caused no change and is the least reactive.

KEY POINT

We could also calculate the molar enthalpy change for each reaction – see Chapter 19 to learn how to do this.

KEY POINT

The general trends on this page apply to simple acids such as dilute sulfuric acid or dilute hydrochloric acid. Nitric acid and concentrated sulfuric acid behave completely differently with most metals. These are problems beyond International GCSE.

A GENERAL SUMMARY

The pattern for the reaction of metals with acids is the same as for the reaction between the metals and water, but in each case the reaction is much more vigorous.

METALS ABOVE HYDROGEN IN THE REACTIVITY SERIES

HINT

You can remember this using the mnemonic MASH.

Metals above hydrogen react with acids to form a salt (e.g. magnesium sulfate or zinc chloride) and hydrogen. The higher the metal in the series, the more violent the reaction.

metal + acid → salt + hydrogen

metal + dilute sulfuric acid → metal sulfate + hydrogen

metal + dilute hydrochloric acid → metal chloride + hydrogen

METALS BELOW HYDROGEN IN THE REACTIVITY SERIES

Metals such as copper, silver and gold do not react with simple dilute acids such as sulfuric or hydrochloric acid.

POTASSIUM, SODIUM, LITHIUM AND CALCIUM WITH DILUTE ACIDS

These are too reactive to add safely to acids, the reaction is too violent. Calcium can be used if the acid is very dilute.

METALS FROM MAGNESIUM TO IRON IN THE REACTIVITY SERIES

!

Safety Note: Wear eye protection and avoid splashing the acid when stirring.

Magnesium reacts vigorously with cold dilute acids, and the mixture becomes hot. A colourless solution of magnesium sulfate or chloride is formed. With dilute sulfuric acid:

$$Mg(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2(g)$$

Aluminium is slow to start reacting, but after warming it reacts very vigorously. There is a very thin, but very strong, layer of aluminium oxide on the surface of the aluminium, which stops the acid from getting to it. On heating, the acid removes this layer, and the aluminium can show its true reactivity. With dilute hydrochloric acid:

$$2Al(s) + 6HCl(aq) \rightarrow 2AlCl_3(aq) + 3H_2(g)$$

Zinc and ***iron*** react slowly in the cold, but more rapidly on heating. The vigour of the reactions is less than that of aluminium. The zinc forms zinc sulfate or zinc chloride and hydrogen. The iron forms iron(II) sulfate or iron(II) chloride and hydrogen. For example:

$$Zn(s) + H_2SO_4(aq) \rightarrow ZnSO_4(aq) + H_2(g)$$
$$Fe(s) + 2HCl(aq) \rightarrow FeCl_2(aq) + H_2(g)$$

KEY POINT

There are two ranges of iron salts containing either Fe^{2+} or Fe^{3+} ions. These reactions produce iron(II) salts, containing Fe^{2+} ions.



▲ Figure 14.15 Iron reacting with dilute hydrochloric acid

FINDING THE APPROXIMATE POSITION OF A METAL IN THE REACTIVITY SERIES USING WATER AND DILUTE ACIDS

KEY POINT

It would be impossible to place a metal exactly in the reactivity series using this sequence. How vigorously a metal reacts with a dilute acid depends on its position in the series, but it also depends on its surface area, and whether the surface is free from dirt or an oxide coating.

Add a very small piece of metal to some cold water. If there is any rapid reaction, then the metal must be above magnesium in the reactivity series.

If there isn't any reaction, add a small amount of metal to some dilute hydrochloric acid (or dilute sulfuric acid). If there isn't any reaction in the cold acid, warm it carefully.

If there is still no reaction, the metal is probably below hydrogen in the reactivity series. If there is a reaction, then it is somewhere between magnesium and hydrogen.

MAKING PREDICTIONS USING THE REACTIVITY SERIES

You can make predictions about the reactions of unfamiliar metals if you know their position in the reactivity series.

A PROBLEM INVOLVING MANGANESE

KEY POINT

You need to know the charge on the ion so that you can deduce the formulae of the manganese compounds. If you aren't sure about working out the formula of an ionic compound, look again at Chapter 7.

Suppose you have a question as follows.

Manganese (Mn) lies between aluminium and zinc in the reactivity series, and forms a 2+ ion. Solutions of manganese(II) salts are very pale pink (almost colourless).

a Use the reactivity series to predict whether manganese will react with copper(II) sulfate solution. If it will react, describe what you would see, name the products and write an equation for the reaction.

b Explain why you would expect manganese to react with steam. Name the products of the reaction and write the equation.

potassium
sodium
lithium
calcium
magnesium
aluminium
manganese
zinc
iron
(hydrogen)
copper

▲ Figure 14.16 Where is manganese in the reactivity series?

a The reaction between manganese and copper(II) sulfate solution

Manganese is above copper in the reactivity series and so will displace it from the copper(II) sulfate:

A pink-brown deposit of copper will be formed. The colour of the solution will fade from blue and leave a very pale pink (virtually colourless) solution of manganese(II) sulfate.

$$Mn(s) + CuSO_4(aq) \rightarrow MnSO_4(aq) + Cu(s)$$

b The reaction between manganese and steam

Manganese is above hydrogen in the reactivity series and so reacts with steam to give hydrogen and the metal oxide, in this case manganese(II) oxide.

You couldn't predict the colour of the manganese(II) oxide, and the question doesn't ask you to do this.

$$Mn(s) + H_2O(g) \rightarrow MnO(s) + H_2(g)$$

RUSTING OF IRON

HINT

Warning! Many metals corrode, but it is only the corrosion of iron that is referred to as *rusting*.

Iron **rusts** in the presence of *oxygen* and *water*. Rusting occurs with iron and the most common alloy of iron, mild steel (see below).

The formula of rust is $Fe_2O_3 \cdot xH_2O$, where x is a variable number. It can be called 'hydrated iron(III) oxide'. Rusting can be described as a redox reaction – the iron is oxidised.

KEY POINT

Forming rust from iron is a surprisingly complicated process. The iron loses electrons to form iron(II) ions, Fe^{2+}, which are then oxidised by the air to iron(III) ions, Fe^{3+}. Reactions involving the water produce the actual rust.



▲ Figure 14.17 Rusting is accelerated by salty water.

PREVENTING RUSTING BY USING BARRIERS

The most obvious way of preventing rusting is to keep water and oxygen away from the iron. You can do this by painting it, coating it in oil or grease, or covering it in plastic. Coating the iron with a metal *below* it in the reactivity series (e.g. coating steel with tin for tin cans) is also a **barrier method**.

Barrier methods are usually quite cheap ways of preventing rusting. A problem with barrier methods is that once the coating is broken, the iron underneath is exposed to oxygen and water and the iron will rust (even the bits that are not directly exposed to the air/water).

PREVENTING RUSTING BY GALVANISING

Galvanised iron is iron that is coated with a layer of zinc. As long as the zinc layer is unscratched, it serves as a barrier to air and water. However, the iron still doesn't rust, even when some of the zinc on the surface is scratched away

EXTENSION WORK

During the process, the zinc loses electrons (is oxidised) to form zinc ions.

$$Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$$

These electrons flow into the iron. Any iron atom which has lost electrons to form an ion immediately regains them. If the iron can't form ions, it can't rust.

to expose the iron. This is because the zinc is more reactive than iron, and so reacts with oxygen/water more readily than the iron does. The zinc corrodes instead of the iron.



▲ Figure 14.18 Galvanised iron doesn't rust even in constant contact with air and water.

PREVENTING RUSTING BY USING SACRIFICIAL PROTECTION



▲ Figure 14.19 A sacrificial anode on a ship

Zinc, magnesium or aluminium blocks are attached to metal hulls or keels of ships to prevent the iron/steel from rusting. For this to work, you have to use a metal that is *more reactive than iron;* the more reactive metal reacts (is oxidised) more readily in the presence of oxygen/water than the iron. The corrosion of the more reactive metal prevents the iron from rusting. Such blocks are called sacrificial anodes.

These sacrificial anodes have to be replaced occasionally when all the more reactive metal has been oxidised. This type of protection is used on large structures where it would be very difficult to use a barrier method effectively.

Galvanising is a combination of a barrier method and **sacrificial protection**.

Underground pipelines are also protected using sacrificial anodes. In this case, lumps of magnesium are attached at intervals along the pipe. The very reactive magnesium corrodes in preference to the iron. The electrons produced as the magnesium forms its ions prevent the ionisation of the iron.

CHAPTER QUESTIONS

SKILLS CRITICAL THINKING (6)

1 a List the following metals in order of decreasing reactivity: aluminium, copper, iron, sodium.

b Some magnesium powder was mixed with some copper(II) oxide and heated strongly. There was a vigorous reaction, producing a lot of sparks and a bright flash of light.

(5) i State the names of the products of the reaction.

SKILLS PROBLEM SOLVING (9) ii Write a balanced symbol equation for the reaction.

SKILLS REASONING iii Explain which substance in the reaction has been reduced.

iv Explain which substance is the oxidising agent.

c If a mixture of zinc powder and cobalt(II) oxide is heated, the following reaction occurs:

$$Zn(s) + CoO(s) \rightarrow ZnO(s) + Co(s)$$

(7) i Explain which metal is higher in the reactivity series.

SKILLS CRITICAL THINKING (9) ii The zinc can be described as a reducing agent. Using this example, explain what is meant by the term 'reducing agent'.

iii Explain which substance in this reaction has been oxidised.

SKILLS REASONING

8

d Aluminium, chromium and manganese are all moderately reactive metals. Use the following information to arrange them in the correct reactivity series order. Start with the most reactive one.

- Chromium is manufactured by heating chromium(III) oxide with aluminium.
- If manganese is heated with aluminium oxide there is no reaction.
- If manganese is heated with chromium(III) oxide, chromium is produced.

10

2 Study the following equations and in each case decide whether the substance in bold type is oxidised or reduced. Explain your choice in terms of either oxygen transfer or electron transfer as appropriate.

a **Zn(s)** + CuO(s) → ZnO(s) + Cu(s)

b **Fe_2O_3(s)** + 3C(s) → 2Fe(s) + 3CO(g)

c **Mg(s)** + Zn^{2+}(aq) → Mg^{2+}(aq) + Zn(s)

d Zn(s) + **Cu^{2+}(aq)** → Zn^{2+}(aq) + Cu(s)

8

3 The equation for the reaction when solid magnesium and solid lead(II) oxide are heated together is:

Mg(s)+ PbO(s) → MgO(s) + Pb(s)

Explain what this tells you about the position of lead in the reactivity series.

6

4 Some iron filings were shaken with some copper(II) sulfate solution. The ionic equation for the reaction is:

Fe(s) + Cu^{2+}(aq) → Fe^{2+}(aq) + Cu(s)

a Explain any one change that you would observe during this reaction.

9

b Explain which substance has been oxidised in this reaction.

SKILLS CRITICAL THINKING

c Write down the full (not ionic) equation for this reaction.

SKILLS REASONING

8

5 Some experiments were carried out to place the metals copper, nickel and silver in reactivity series order.

Experiment 1: A piece of copper was placed in some green nickel(II) sulfate solution. There was no change to either the copper or the solution.

Experiment 2: A coil of copper wire was suspended in some silver nitrate solution. A furry grey growth appeared on the copper wire, out of which grew spiky silvery crystals. The solution gradually turned from colourless to blue.

a Use this information to place copper, nickel and silver in reactivity series order, starting with the most reactive one.

b In another experiment, a piece of nickel was placed in some copper(II) sulfate solution.

6

i State one change that you would observe during this reaction.

SKILLS PROBLEM SOLVING

9

ii Write the full balanced equation for this reaction.

iii Write the ionic equation for this reaction and use it to explain which substance has been oxidised during the reaction.

SKILLS REASONING

8

6 a Look carefully at the following equations and then explain what you can say about the position of the metal X in the reactivity series.

X(s) + 2HCl(aq) → XCl_2(aq) + H_2(g)

X(s) + $CuSO_4$(aq) → XSO_4(aq) + Cu(s)

X(s) + $FeSO_4$(aq) → no reaction

SKILLS REASONING 8

b Decide whether X will react with the following substances. If it will react, state the names of the products, if not state 'no reaction'.

i Silver nitrate solution
ii Zinc oxide
iii Cold water
iv Copper(II) chloride solution
v Dilute sulfuric acid

SKILLS CRITICAL THINKING 5

7 If you add some powdered aluminium to a small amount of cold dilute hydrochloric acid in a boiling tube, very little happens. If you warm this gently, it starts to fizz very rapidly.

a State the name of the gas given off to produce the fizzing.

b If you used an excess of hydrochloric acid, the result would be a colourless solution. State the name of the solution.

SKILLS PROBLEM SOLVING 9

c Write the full balanced equation for the reaction.

SKILLS REASONING 6

d Explain why the aluminium hardly reacts at all with the dilute acid in the cold, but reacts vigorously after even gentle heating.

SKILLS EXECUTIVE FUNCTION 8

8 If you have some small pieces of the metal titanium and any simple apparatus that you might need, describe how you would find out the approximate position of titanium in the reactivity series, using only water and dilute hydrochloric acid. You only need to find out that the reactivity is 'similar to iron' or 'similar to magnesium', for example. Your experiments should be done in an order that guarantees maximum safety. For example, if titanium's reactivity turned out to be similar to potassium, dropping it into dilute hydrochloric acid wouldn't be a good idea!

SKILLS CRITICAL THINKING 5

9 In the past, cars were made from mild steel, which was then painted. In more modern cars, the mild steel is galvanised before it is painted.

a State the conditions under which iron/steel rusts.

6

b Explain how painting prevents iron/steel from rusting.

c What is meant by *galvanised steel*?

7

d Describe and explain the effect that galvanised steel has on the life of the car.

SKILLS INTERPRETATION 8

10 A student tested four unknown metals (the names are all imaginary) with different metal salt solutions. She put a tick (✔) if there was a reaction and a cross (✖) if there was no change. Some of her results are shown in the table.

	Pearsonium chloride solution	Mollium chloride solution	Rosium chloride solution	Amelium chloride solution
pearsonium	✖		✖	
mollium	✔	✖	✔	✔
rosium			✖	✖
amelium				✖

a Copy and complete the table.

SKILLS CRITICAL THINKING

b Arrange the metals in an order of reactivity (most reactive first).

SKILLS PROBLEM SOLVING 6

c Copy and complete the equation:

mollium + pearsonium chloride →